TAKE-HOME ASSESSMENT

Medical Scheduling System

Multi-party appointment scheduling with optimal slot finding

Made with GAMMA

The Challenge

Healthcare scheduling requires coordinating multiple parties while avoiding conflicts. Our system solves three core problems:

Multi-Party Coordination

Schedule Doctor, NMT, and Scan resources simultaneously

Optimal Slot Finding

Locate first available continuous time window

Conflict Prevention

Eliminate overlaps and double-bookings

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System Architecture

Client-server design with a monolithic backend for simplicity and rapid development.

Angular 18

Single Page Application

FastAPI

REST API Layer

SQLite

Persistent Storage

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Key Features

Visual Availability Grid

36-slot display covering 8AM–5PM in 15-minute intervals

Care Path Builder

Define appointment sequences with gap support between steps

JWT Authentication

Secure token-based access control

Interactive Controls

Drag-select to block/unblock slots with real-time highlighting

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CORE ALGORITHM

NumPy-Powered Slot Finding

01

Build 2D Matrix

Rows = parties, Columns = slots

02

Apply Mask Matrix

Map care path requirements

03

Sliding Window Check

Detect cells = 2 (conflict)

04

Return Result

First valid position found

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Design Assumptions

Patient Availability

Patients are always available—blocked by default in the scheduling matrix

Fixed Time Slots

36 slots per day (8AM–5PM) with 15-minute granularity

Single-Day Scope

Scheduling operates within one calendar day at a time

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Technology Stack

Layer	Technology	Purpose
Frontend	Angular 18, TypeScript	Interactive UI
Backend	FastAPI, Python	REST API
Database	SQLite, SQLAlchemy	Data persistence
Auth	JWT	Secure sessions
Algorithm	NumPy	Matrix operations

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Future Enhancements

Administration

- Dynamic party management
- Role-based access control
- Analytics dashboard

Patient Experience

- Patient information storage
- Notifications & reminders
- Multi-day scheduling

INTERACTIVE

Live Demo

See the scheduling system in action—from availability grid to optimal slot finding.

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Thank You

Questions or feedback welcome.

36	O(n)	5
Time Slots	Algorithm	Tech Layers

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